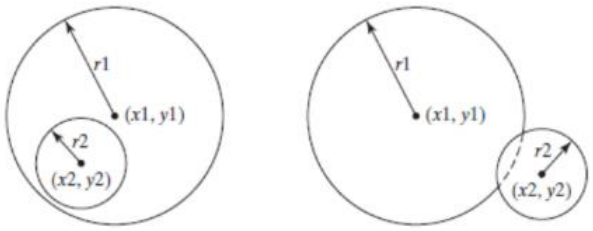
Week 2 Exercises

Try as many of these exercises as you can before the seminar. In the seminar we will work through a selection of them and discuss your solutions, so bring your questions and anything you got stuck on. The exercises are grouped by topic, and within each group they go from easier to harder.

Conditions and loops

1. (**Geometry: point in a circle?**) Write a program that prompts the user to enter a point (x, y) and checks whether the point is within the circle centered at (0, 0) with radius 10. For example, (4, 5) is inside the circle and (9, 9) is outside the circle. Test your program to cover all cases.

2. (**Geometry: two circles**) Write a program that prompts the user to enter the center coordinates and radii of two circles and determines whether the second circle is inside the first or overlaps with the first, as shown in Figure:

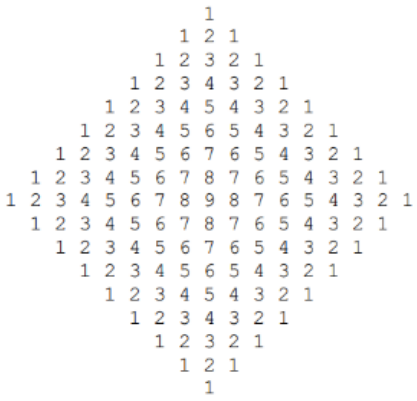


(Hint: circle2 is inside circle1 if the distance between the two centers <= |r1 -r2| and circle2 overlaps circle1 if the distance between the two centers <= r1 + r2.) Test your program to cover all cases.

3. (**Finance: compute future tuition**) Suppose that the tuition for a university is 400,000 lekë this year and increases 5% every year. In one year, the tuition will be 420,000 lekë. Write a program that computes the tuition in ten years and the total cost of four years' worth of tuition after the tenth year.

4. **(Game: craps)** Craps is a popular dice game played in casinos. Write a program to play a variation of the game, as follows: Roll two dice. Each die has six faces representing the values 1 to 6. Add the values of the two dice. If the sum is 2, 3, or 12 (called craps), you lose. If the sum is 7 or 11 (called natural), you win. If the sum is any other value (i.e., 4, 5, 6, 8, 9, or 10), that value becomes the point. In that case, keep rolling the dice until either a 7 or the point value is rolled. If a 7 is rolled, you lose. If the point value is rolled, you win.

5. (**Draw Full Diamond**) Write a code with nested loops to display the following pattern:



6. (**Compute e)** You can approximate e using the following series:

$$e = 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \ldots + \frac{1}{i!}$$

Write a program that displays the e value for i = 10000, 20000, ..., and 100000.

Methods

1. **(Conversions between Celsius and Fahrenheit)** Write a class that contains the following two methods:

/** Convert from Celsius to Fahrenheit */

public static double celsiusToFahrenheit (double celsius)

/** Convert from Fahrenheit to Celsius */

public static double fahrenheitToCelsius(double fahrenheit)

The formulas for the conversion are:

fahrenheit = (9.0 / 5) * celsius + 32

celsius = (5.0 / 9) * (fahrenheit – 32)

Write a test program that invokes these methods.

2. **(Math: pentagonal numbers)** A pentagonal number is defined as n(3n–1)/2 for n = 1, 2, 3 . . ., and so on. Therefore, the first few numbers are 1, 5, 12, 22, ... Write a method with the following header that returns a pentagonal number:

public static int getPentagonalNumber(int n)

Write a test program that uses this method to display the first 100 pentagonal numbers with 10 numbers on each line.

3. (**Financial application: calculate interest**) If you know the balance and the annual percentage interest rate, you can compute the interest on the next monthly payment using the following formula:

interest = balance * (annualInterestRate / 1200)

a) Write a program that reads the balance and the annual percentage interest rate and displays the interest for the next month.

b) Rewrite the program using input and output dialog boxes (JOptionPane).

4. (**Average an array**) Write two overloaded methods that return the average of an array with the following headers:

public static int average(int[] array)

public static double average(double[] array)

Write a test program that prompts the user to enter ten double values, invokes this method, and displays the average value.

5. (**Arguments: Min and Max**) Write two functions that get undetermined number of parameters to find the minimum or the maximum

public static int max(int... n)

public static int min(int... n)

Write a test program that displays the maximum and the minimum of a sequence of numbers

Arrays

1. **(Reverse the numbers entered**) Write a program that reads ten integers and displays them in the reverse of the order in which they were read.

2. (**identical arrays**) The arrays list1 and list2 are identical if their corresponding elements are equal. Write a method that returns true if list1 and list2 are identical, using the following header:

public static boolean identical(int[] list1, int[] list2)

Write a test program that prompts the user to enter two lists of integers and displays whether the two are identical. The first number in the input should indicate the number of the elements in the list.

Here is a sample run:

```
Enter list1: 5 2 5 6 1 6 ↵Enter
Enter list2: 5 2 5 6 1 6 ↵Enter
Two lists are identical
```

3. (**Largest row and column**) Write a program that randomly fills in 0s and 1s into a 4-by-4 matrix, prints the matrix, and finds the first row and column with the most 1s. Here is a sample run of the program:

```
0 0 1 1
0 0 1 1
1 1 0 1
1 0 1 0

The largest row index: 2
The largest column index: 2
```

4. (**Student grade averages**) Create respective arrays for the following data:

	Grade 1	Grade 2	Grade 3	Grade 4	Grade 5
Jack	76	54	89	76	98
Robert	34	65	23	87	100
Edward	80	65	97	54	94
Eddie	63	75	33	75	87

Create a method which returns an array of integers that will hold the average of each student.

public static int[] calculateAverage(int[][] grades)

Create a method that prints all the arrays in a formatted way

public static void printStudents(String[] names, int[][] grades, int[] average)

Create another method which sorts all the arrays according to the average

public static void sort(String[] names, int[][] grades, int[] average)

Test all the methods: Calculate average -> Print the arrays -> Sort the arrays -> Print the arrays again

Strings and characters

1. (**Generate vehicle plate numbers**) Assume a vehicle plate number consists of three uppercase letters followed by four digits. Write a program to generate a plate number.

2. (**Checking password**) Some Websites impose certain rules for passwords. Write a method that checks whether a string is a valid password. Suppose the password rule is as follows:

o A password must have at least eight characters.

o A password consists of only letters and digits.

o A password must contain at least two digits.

Write a program that prompts the user to enter a password and displays "Valid Password" if the rule is followed or "Invalid Password" otherwise.

3. (**Occurrences of each digit in a string**) Write a method that counts the occurrences of each digit in a string using the following header:

public static int[] count(String s)

The method counts how many times a digit appears in the string. The return value is an array of ten elements, each of which holds the count for a digit. For example, after executing int[] counts = count("12203AB3"), counts[0] is 1, counts[1] is 1, counts[2] is 2, counts[3] is 2.

Write a test program that prompts the user to enter a string and displays the number of occurrences of each digit in the string.

4. (**Binary to decimal**) Write a method that parses a binary number as a string into a decimal integer. The method header is as follows:

public static int binaryToDecimal(String binaryString)

For example, binary string 10001 is 17 So, binaryToDecimal("10001") returns 17. Note that Integer.parseInt("10001", 2) parses a binary string to a decimal value. Do not use this method in this exercise.

Write a test program that prompts the user to enter a binary string and displays the corresponding decimal integer value.